

NTC Hospital Presentation

Delivering Quality Healthcare with
Compassion and Excellence



Presented by Tafadzwa Diana Mavunga



Vision

To be a leading healthcare institution recognized for data-driven clinical excellence, proactive chronic disease management, and superior patient outcomes.

Mission

NTC Hospital is dedicated to providing high-quality, comprehensive healthcare by optimizing our operational resources, standardizing patient care pathways, and implementing advanced care transition strategies to ensure a safe, sustainable recovery for every patient.

Core Components of the Presentation:

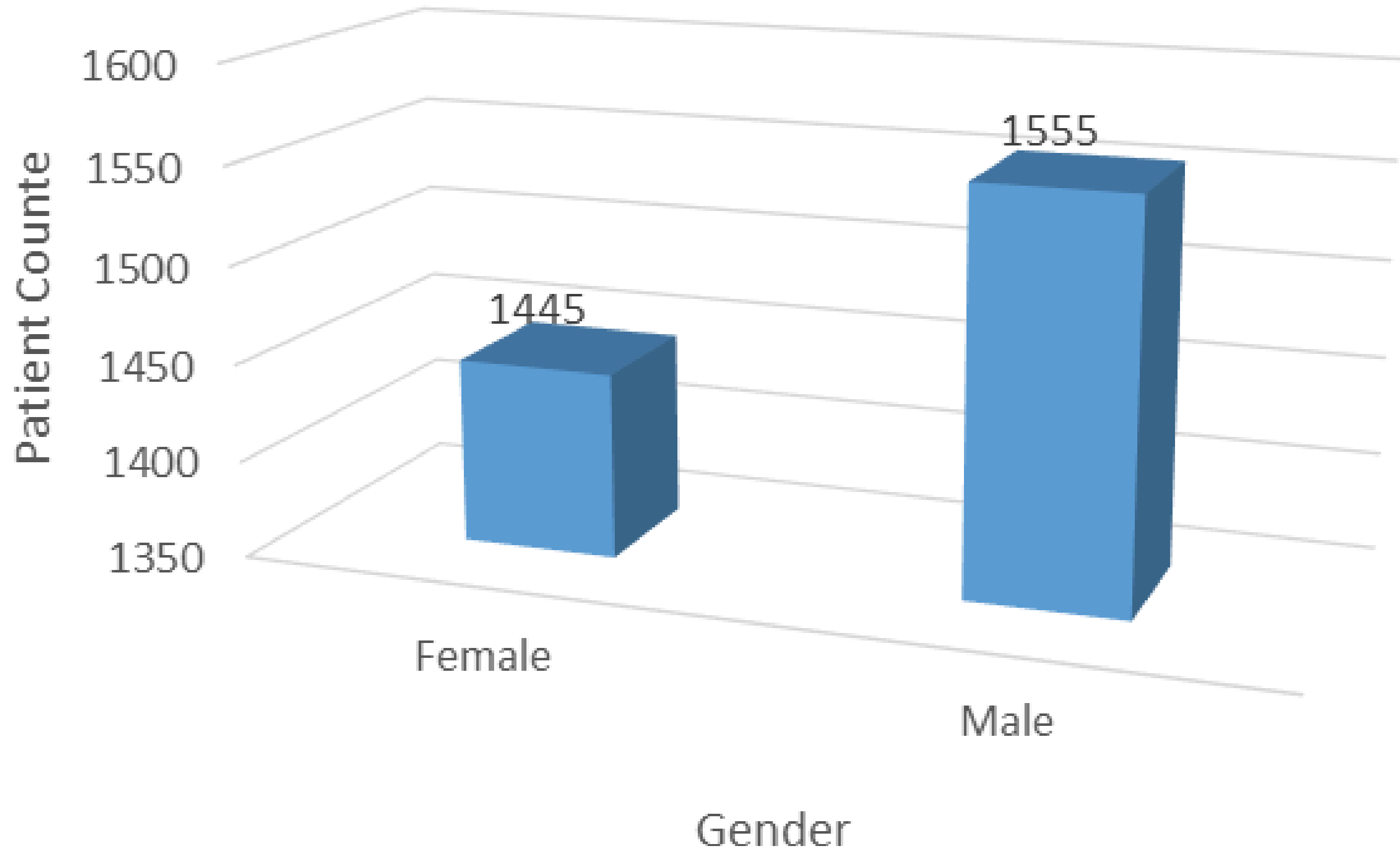
The Clinical Baseline: It analyzes a core dataset of 3,000 unique patient clinical records to identify patterns behind hospital stays and returning patient volume.



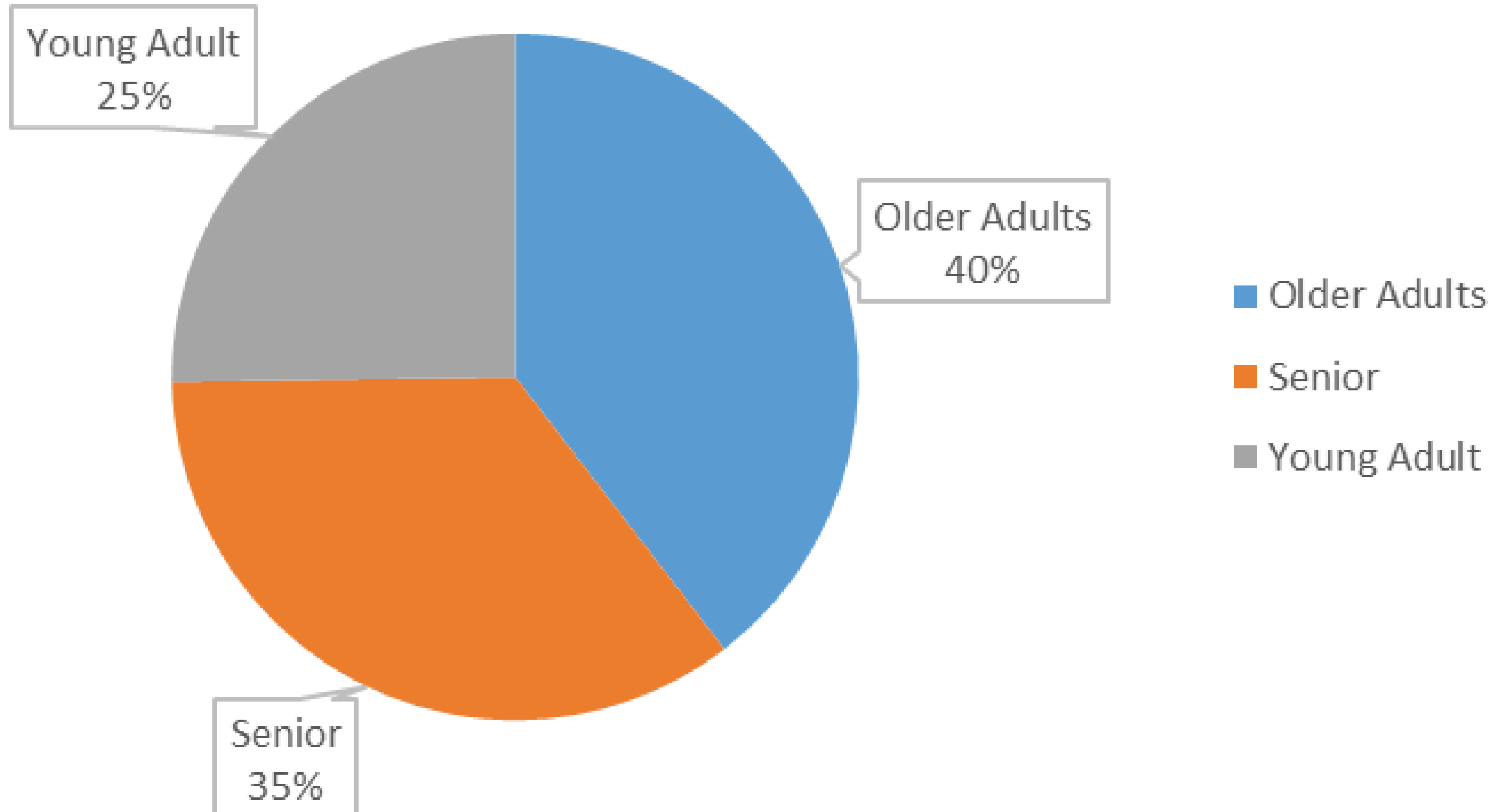
Graphical Data Insights: The slide deck walks through a detailed visual breakdown of patient demographics, tracking metrics like length of stay by age group, diagnosis complexity, blood sugar concentrations, and blood pressure averages.

Strategic Directives: It translates those analytical findings into clinical and operational recommendations—such as advanced care transition protocols for high-risk older adults and institutional chronic disease initiatives—to actively minimize hospital readmission rates.

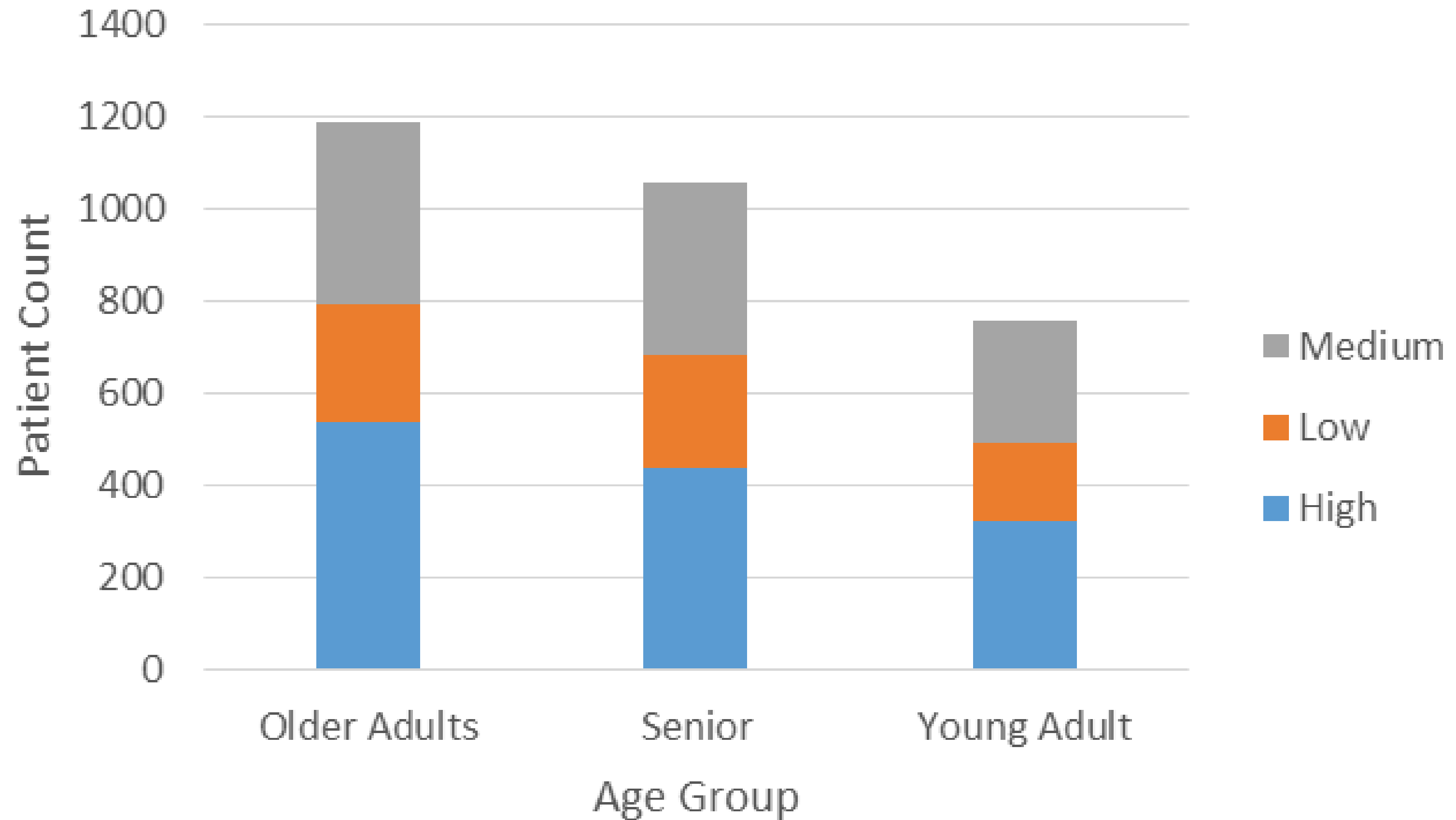
Readmission Count by Gender



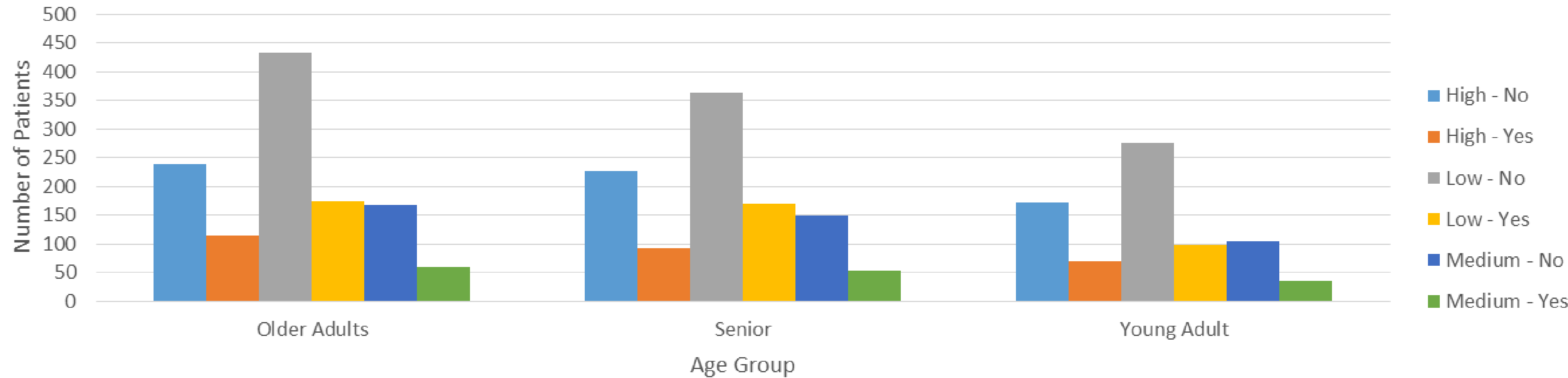
Age_Group by Length of Stay



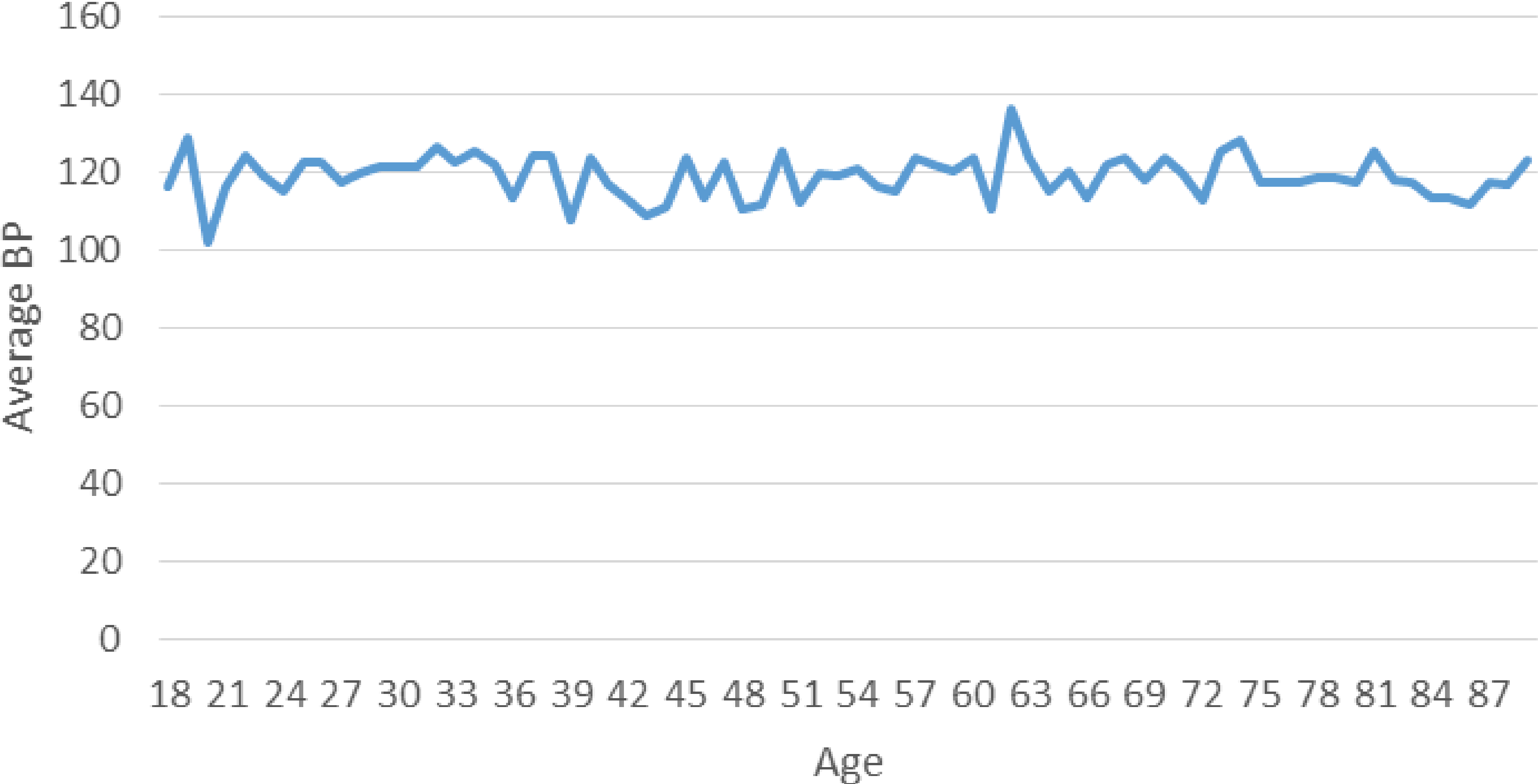
Patient Diagnosis Complexity by Age Cohort



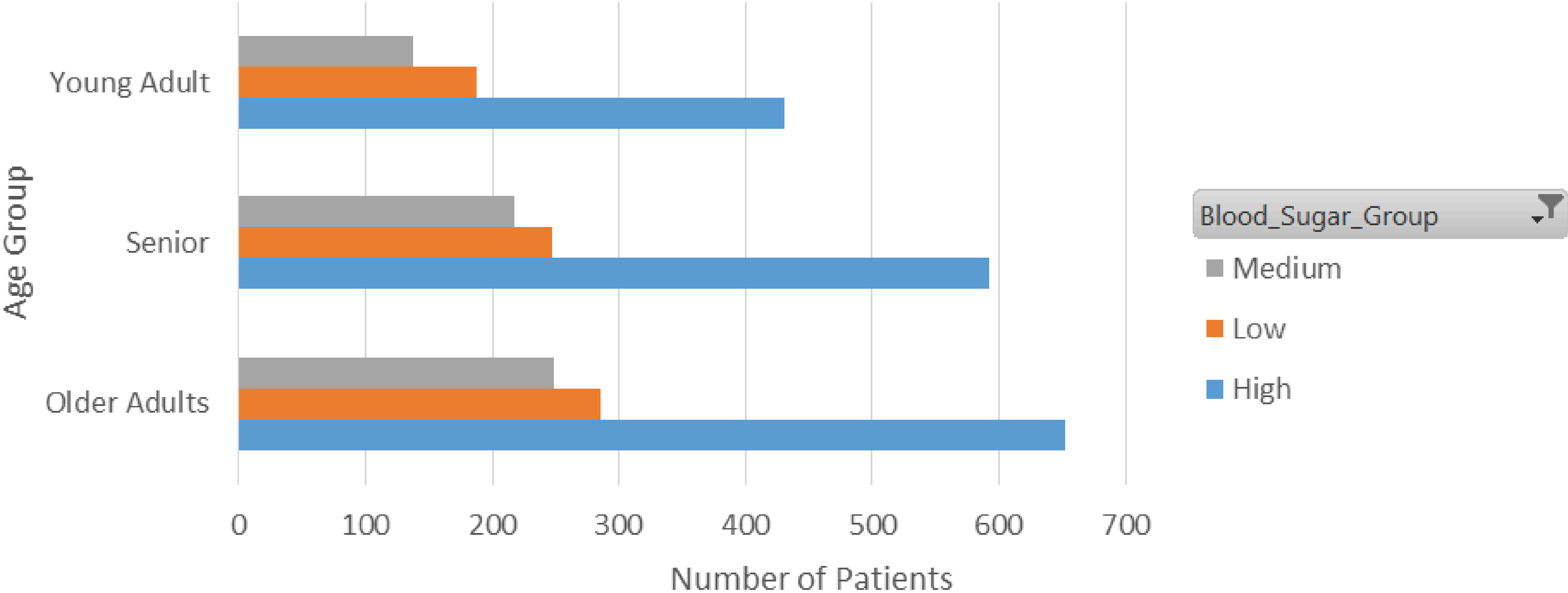
Blood Pressure & Readmission by Age Group



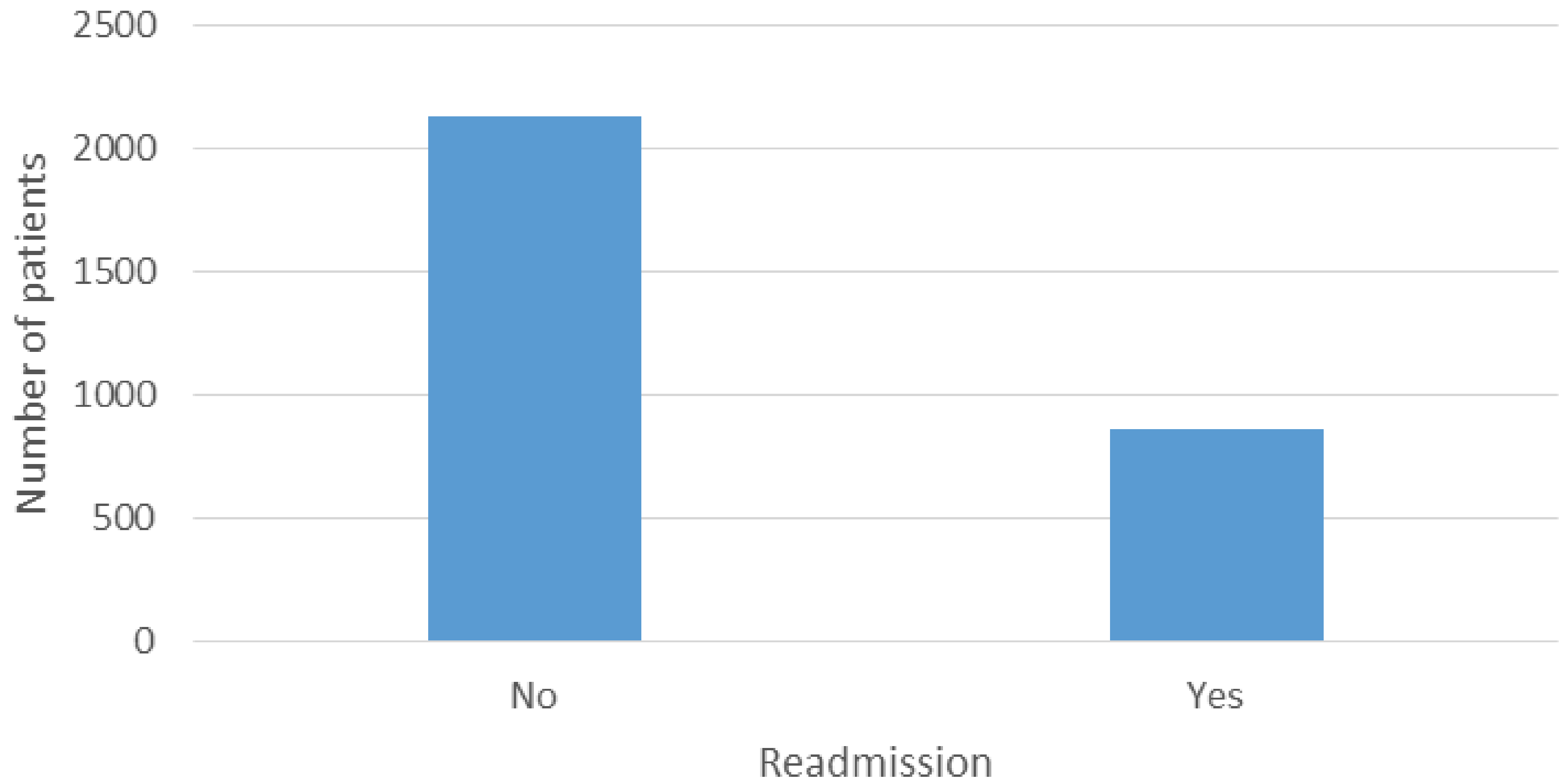
Average BP Per Age



Blood Sugar Levels by Age Group



Patient Readmission Breakdown



Insights:

Gender: Males have higher readmissions (1,555 vs. 1,445), driven by distinct chronic health profiles.

Length of Stay: "Older Adults" (40%) and Seniors (35%) stay longest due to slower baseline recovery rates.

Complexity: High-complexity cases peak in "Older Adults" (540) due to higher rates of overlapping chronic conditions.

BP & Readmissions: High blood pressure triggers frequent readmissions in "Older Adults" by worsening vascular stiffness.

BP Trend: Averages stay flat (100–130 mmHg) across ages, showing that age alone doesn't skew baseline averages.

Blood Sugar: High blood sugar peaks in "Older Adults" because aging increases insulin resistance and lowers beta-cell function.

Readmission Rate: 866 patients were readmitted within 30 days vs. 2,134 successfully discharged, highlighting gaps in transitional care.

Recommendations:

- **Target High-Risk Older Adults:** Focus 48-hour post-discharge calls and home-health checks on Older Adults, who comprise 40% of the length-of-stay demographic and have the highest volume of high-complexity cases.
 - **Establish a Senior Cohort Continuity Plan:** Create a specialized discharge pathway for Seniors to hand over care seamlessly to primary care or geriatric specialists, aiming to prevent increases in baseline readmissions.
 - **Deploy Standardized Care Bundles:** Implement standardized clinical pathways for high-complexity cases across all age groups, including Young Adults who still present high-severity cases.
 - **Optimize Bed and Staff Allocation:** Use length-of-stay data to forecast bed demand and allocate specialized nursing staff to wards handling long-stay patients (Older Adults and Seniors).
 - **Launch an Institutional Diabetes Initiative:** Implement stricter inpatient glycemic protocols and mandatory diabetic educator consultations before discharge to manage widespread high blood sugar.
 - **Integrate Hypertension and Metabolic Screening:** Combine cardiovascular and metabolic tracking checklists during follow-up care for Older Adults facing both high blood pressure and high blood sugar.
 - **Maintain Current Baseline Care for Low-Risk Patients:** Keep standard monitoring protocols for clinically stable, low-risk segments to free up resources for more complex cases.
 - **Form a Readmission Task Force:** Establish a clinical quality committee to target a 10% reduction in the baseline 28.9% readmission rate over the next fiscal quarter.
-

Thank You

Presented by Tafadzwa Diana Mavunga

